

PROJECT : 4 GOLDEN AMI DEPLOYMENT

Configuration EC2 with web server

Create private S3 bucket

Launch multiple EC2 instance from AMI

Verify identical configuration

➤ Launch & Configure EC2

The screenshot shows the AWS Management Console for the EC2 service in the eu-north-1 region. The main content area displays a table of instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4 D. The first instance listed is 'project-webse...' with Instance ID 'i-0d66c3dada630180d', which is in a 'Running' state and has a 't3.micro' instance type. The status check shows 'Initializing'. The availability zone is 'eu-north-1a' and the public IPv4 address is 'ec2-51-20-87-'. The left sidebar shows the navigation menu with options like Dashboard, AWS Global View, Events, and various EC2 services. The top navigation bar includes the AWS logo, search bar, and account information.

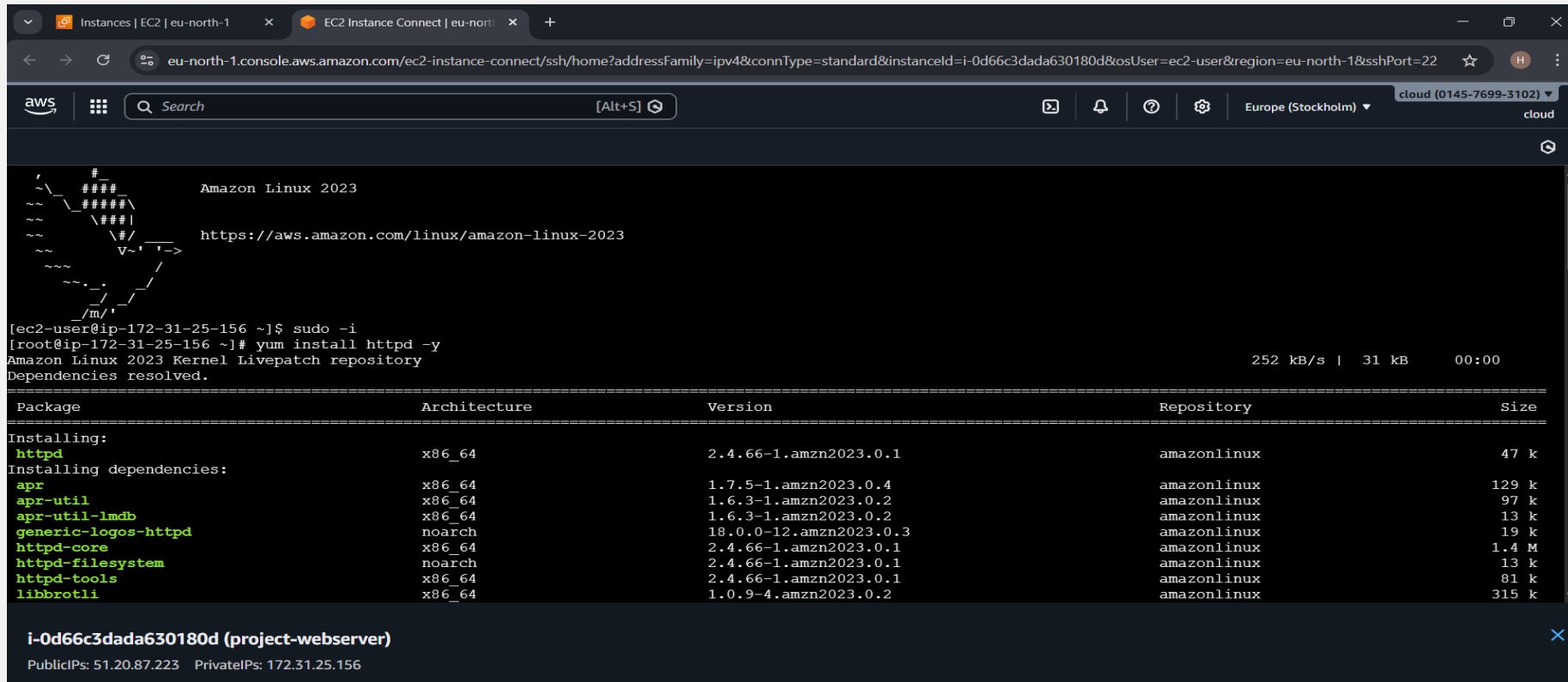
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 D
project-webse...	i-0d66c3dada630180d	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-51-20-87-

➤ Here detail about server project-webserver

The screenshot displays the AWS Management Console interface for an EC2 instance. The browser address bar shows the URL: `eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#InstanceDetails:instanceId=i-0d66c3dada630180d`. The console header includes the AWS logo, a search bar, and navigation links for EC2, Instances, and the specific instance ID `i-0d66c3dada630180d`. The left sidebar lists various EC2 services, with 'Instances' expanded. The main content area is titled 'Instance summary for i-0d66c3dada630180d (project-webserver)' and includes an 'Info' link. It displays the instance's status as 'Running' and provides a comprehensive list of configuration details.

Property	Value
Instance ID	i-0d66c3dada630180d
IPv6 address	-
Hostname type	IP name: ip-172-31-25-156.eu-north-1.compute.internal
Answer private resource DNS name	IPv4 (A)
Auto-assigned IP address	51.20.87.223 [Public IP]
IAM role	-
IMDSv2	Required
Operator	-
Public IPv4 address	51.20.87.223 open address
Instance state	Running
Private IP DNS name (IPv4 only)	ip-172-31-25-156.eu-north-1.compute.internal
Instance type	t3.micro
VPC ID	vpc-0b70c516bf88cd647
Subnet ID	subnet-0f8b0a5b800183eb8
Instance ARN	arn:aws:ec2:eu-north-1:014576993102:instance/i-0d66c3dada630180d
Private IPv4 addresses	172.31.25.156
Public DNS	ec2-51-20-87-223.eu-north-1.compute.amazonaws.com open address
Elastic IP addresses	-
AWS Compute Optimizer finding	Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto Scaling Group name	-
Managed	false

➤ We connect server and install httpd

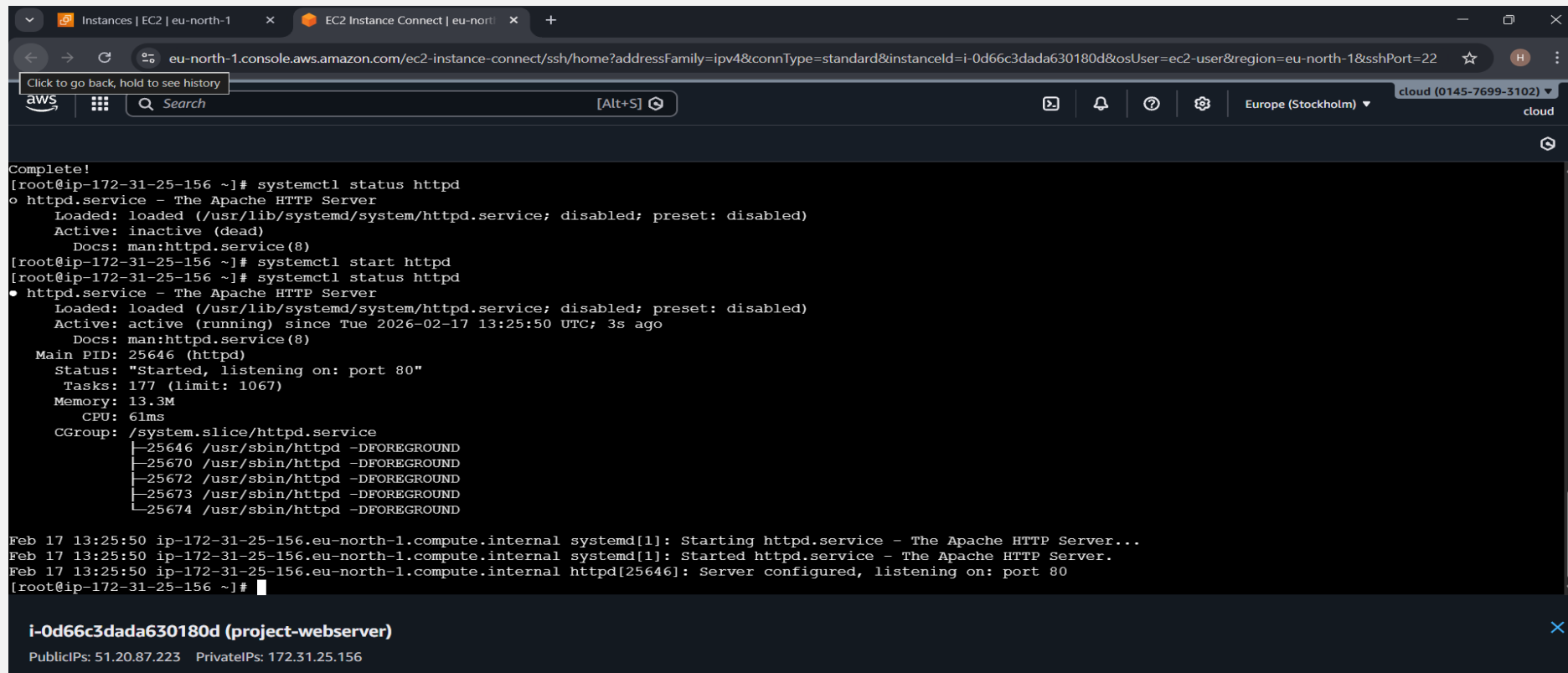


The screenshot shows an AWS console terminal window for an EC2 instance named 'i-0d66c3dada630180d' in the 'eu-north-1' region. The terminal displays the Amazon Linux 2023 logo and the URL 'https://aws.amazon.com/linux/amazon-linux-2023'. The user 'ec2-user' runs 'sudo -i' to become root. Then, 'yum install httpd -y' is executed, which resolves dependencies and shows a table of packages to be installed.

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	47 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	97 k
apr-util-ldap	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	13 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.66-1.amzn2023.0.1	amazonlinux	13 k
httpd-tools	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	81 k
libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k

i-0d66c3dada630180d (project-webserver)
PublicIPs: 51.20.87.223 PrivateIPs: 172.31.25.156

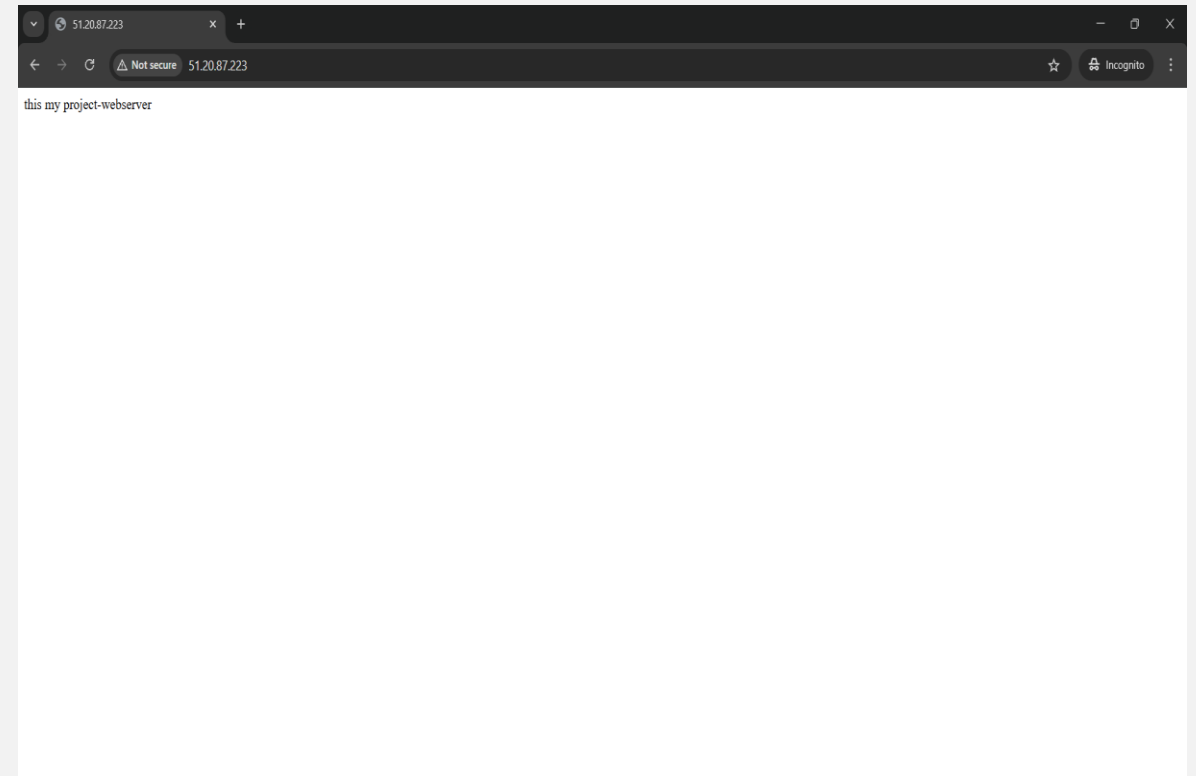
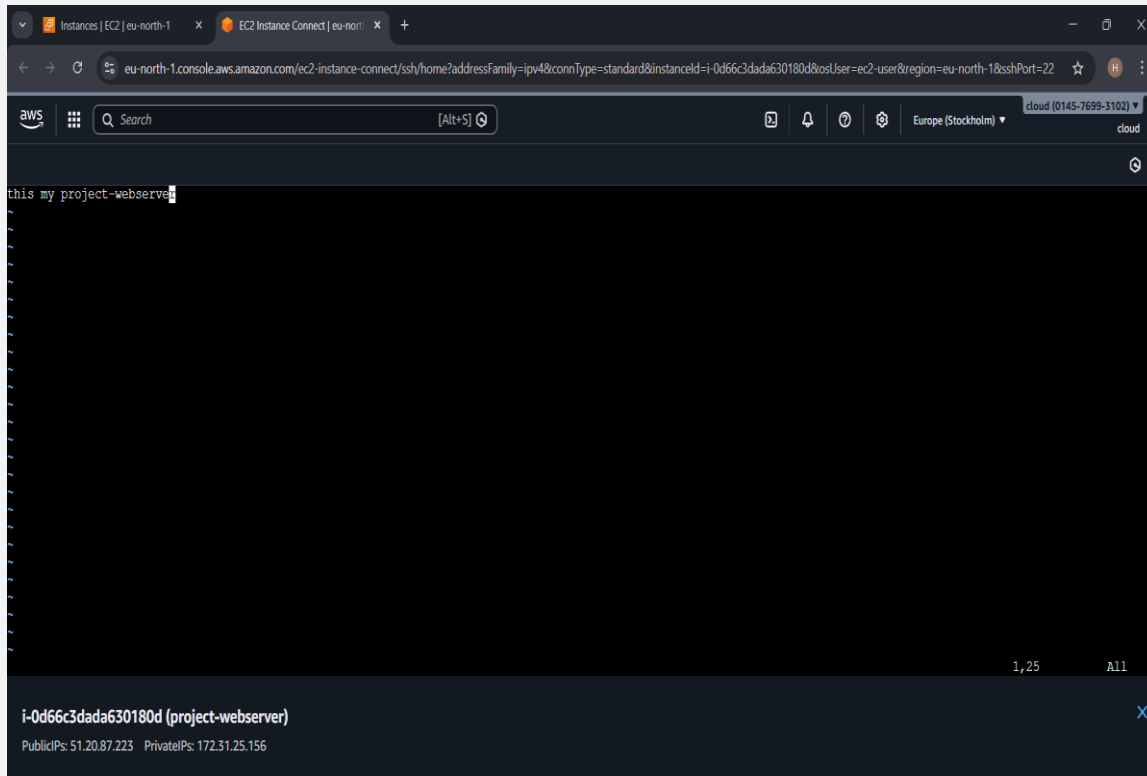
➤ Here we use systemctl to start httpd



```
Complete!  
[root@ip-172-31-25-156 ~]# systemctl status httpd  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)  
   Active: inactive (dead)  
     Docs: man:httpd.service(8)  
[root@ip-172-31-25-156 ~]# systemctl start httpd  
[root@ip-172-31-25-156 ~]# systemctl status httpd  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)  
   Active: active (running) since Tue 2026-02-17 13:25:50 UTC; 3s ago  
     Docs: man:httpd.service(8)  
  Main PID: 25646 (httpd)  
    Status: "Started, listening on: port 80"  
   Tasks: 177 (limit: 1067)  
  Memory: 13.3M  
    CPU: 61ms  
   CGroup: /system.slice/httpd.service  
           └─25646 /usr/sbin/httpd -DFOREGROUND  
           └─25670 /usr/sbin/httpd -DFOREGROUND  
           └─25672 /usr/sbin/httpd -DFOREGROUND  
           └─25673 /usr/sbin/httpd -DFOREGROUND  
           └─25674 /usr/sbin/httpd -DFOREGROUND  
  
Feb 17 13:25:50 ip-172-31-25-156.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...  
Feb 17 13:25:50 ip-172-31-25-156.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.  
Feb 17 13:25:50 ip-172-31-25-156.eu-north-1.compute.internal httpd[25646]: Server configured, listening on: port 80  
[root@ip-172-31-25-156 ~]#
```

i-0d66c3dada630180d (project-webserver)
PublicIPs: 51.20.87.223 PrivateIPs: 172.31.25.156

➤ We make file as index.Html and inside that we write : this my project-website



➤ Create ami with help of previous instance.

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateImage:instanceId=i-0d66c3dada630180d

EC2 > Instances > i-0d66c3dada630180d > Create image

Create image [Info](#)

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Image details

Instance ID
i-0d66c3dada630180d (project-webserver)

Image name
AMI-webserver
Maximum 127 characters. Can't be modified after creation.

Image description - optional
Image description
Maximum 255 characters

☒ **Reboot instance**
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

Storage type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
EBS	/dev/...	Create new snapshot fro...	8	EBS General Purpose SS...	3000		☒ Enable	☐ Enable

➤ Here we can see our ami will be available to create instance.

The screenshot displays the AWS Management Console interface for the eu-north-1 region. The left-hand navigation pane is open, showing the 'EC2' section with a sub-menu for 'Images'. The main content area is titled 'Amazon Machine Images (AMIs) (1)' and includes a search bar and a table of available AMIs. The table lists one AMI with the ID 'ami-07fe6c04faf72a082', sourced from '014576993102/AMI-webserver', owned by '014576993102', with a visibility of 'Private', a status of 'Available', and a creation date of '2026/02/17 19:00 GMT+5:30'. Below the table, there is a section titled 'Select an AMI'.

Amazon Machine Images (AMIs) (1) Info

Owned by me ▼ Find AMI by attribute or tag

AMI ID	Source	Owner	Visibility	Status	Creation date
ami-07fe6c04faf72a082	014576993102/AMI-webserver	014576993102	Private	Available	2026/02/17 19:00 GMT+5:30

Select an AMI

➤ Here we take ami to create multiple EC2 instance

The screenshot displays the AWS Management Console interface for launching an EC2 instance. The browser address bar shows the URL: `eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:`. The console header includes the AWS logo, a search bar, and navigation links for EC2, Instances, and Launch an instance. The main content area is titled "Application and OS Images (Amazon Machine Image)" and includes a search bar and filters for "Recents", "My AMIs", and "Quick Start". Under "My AMIs", the "Owned by me" filter is selected. A list of AMIs is shown, with the "AMI-webserver" (ami-07fe6c04faf72a082) selected. The right sidebar contains a "Summary" section with the following configuration details:

- Number of instances:** 2
- Software Image (AMI):** AMI-webserver (ami-07fe6c04faf72a082)
- Virtual server type (instance type):** t3.micro
- Firewall (security group):** New security group
- Storage (volumes):** 1 volume(s) - 8 GiB

At the bottom right, there are buttons for "Cancel", "Launch instance", and "Preview code".

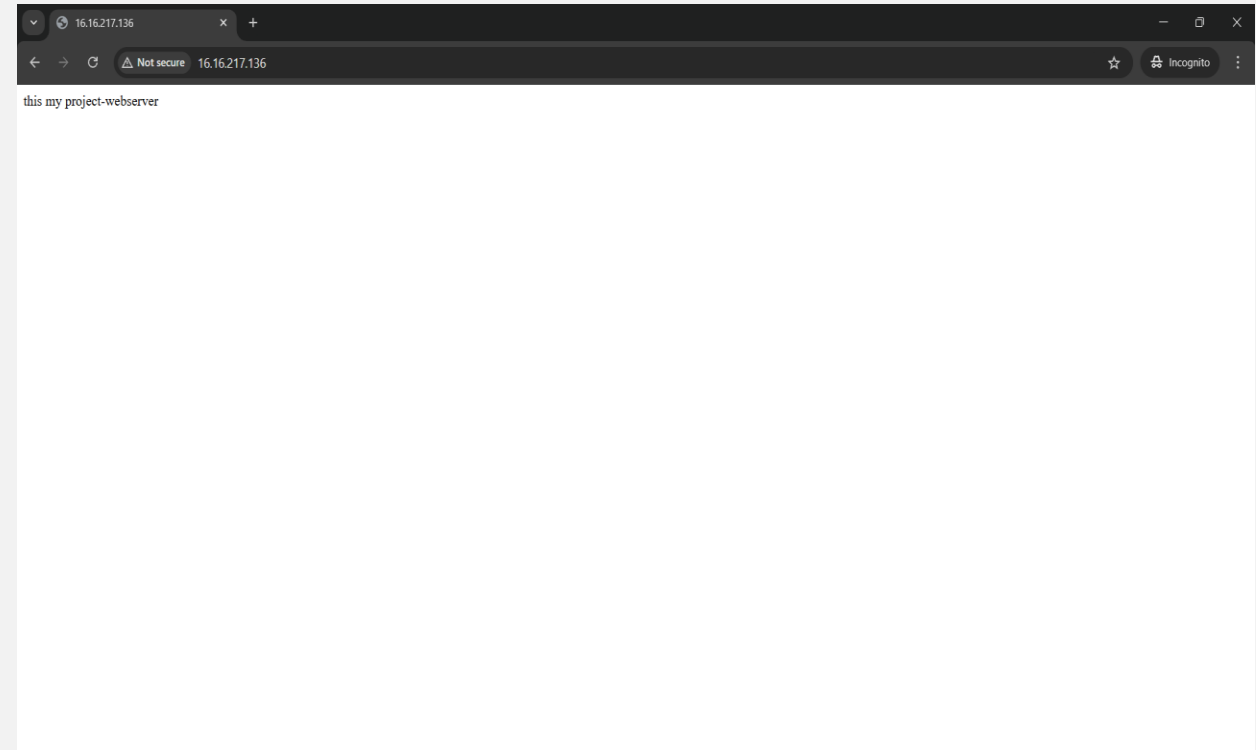
➤ Webserver 1 and webserver 2 are created by using ami

The screenshot displays the AWS Management Console for the 'eu-north-1' region. The left sidebar shows the navigation menu with 'EC2' selected, and 'Instances' is the active view. The main content area shows a list of EC2 instances. Two instances, 'webserver1' and 'webserver2', are selected. Below the list, the 'Monitoring' tab is active, showing four metrics: CPU utilization (%), Network in (bytes), Network out (bytes), and Network packets in (count). All metrics show 'No data available.'.

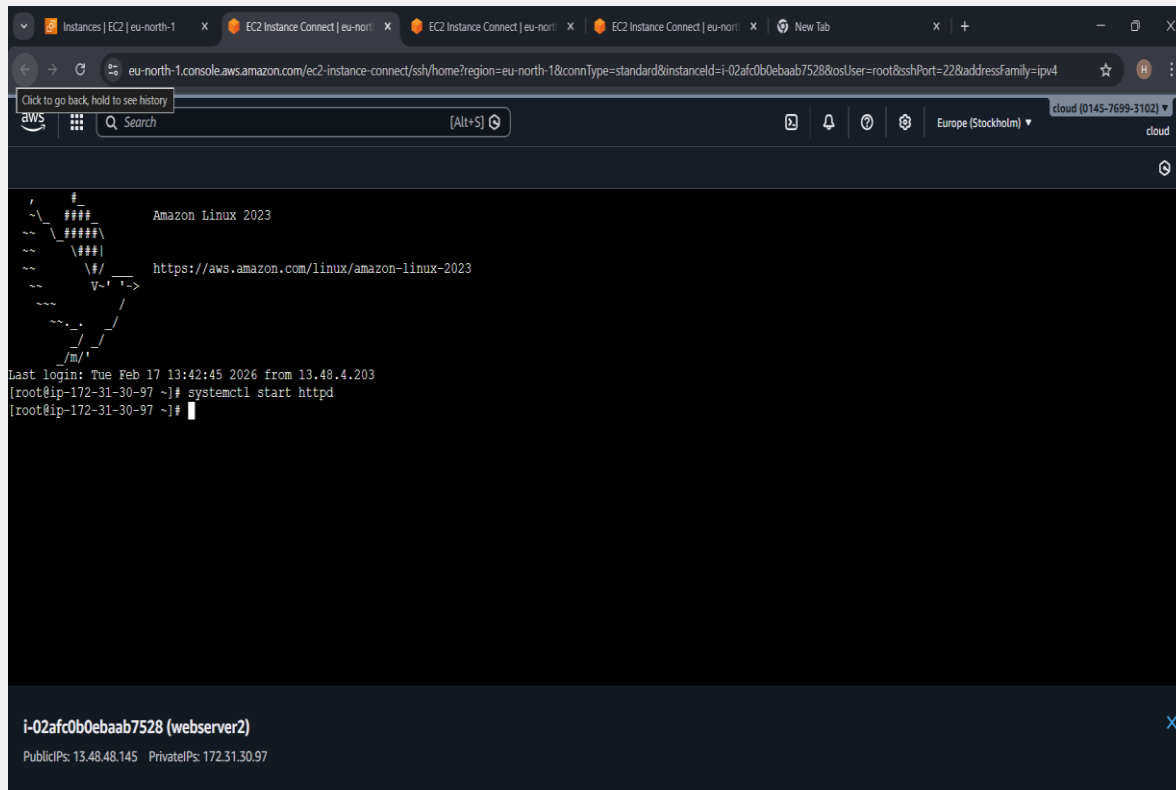
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 D
webserver1	i-0b7231afea5dd73a6	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-16-16-217
webserver2	i-02afc0b0ebaab7528	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-13-48-48-
project-webse...	i-0d66c3dada630180d	Running	t3.micro	3/3 checks pass	View alarms +	eu-north-1a	ec2-51-20-87-

- We connect webserver 1 and start httpd and use public ip to display output

The image shows a web browser window with the AWS Management Console. The address bar displays the URL: eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-0b7231afea5dd73a6&osUser=root&sshPort=22&addressFamily=ipv4. The browser's top bar shows the AWS logo, a search bar, and the region 'Europe (Stockholm)'. The main content area shows a terminal window for an Amazon Linux 2023 instance. The terminal displays the Amazon Linux logo and the URL https://aws.amazon.com/linux/amazon-linux-2023. Below the logo, the command 'systemctl start httpd' is entered and executed. The terminal output shows the command being run and the resulting output. The browser's bottom bar shows the instance ID 'i-0b7231afea5dd73a6 (webserver1)' and the public and private IP addresses: 'PublicIPs: 16.16.217.136 PrivateIPs: 172.31.18.250'.



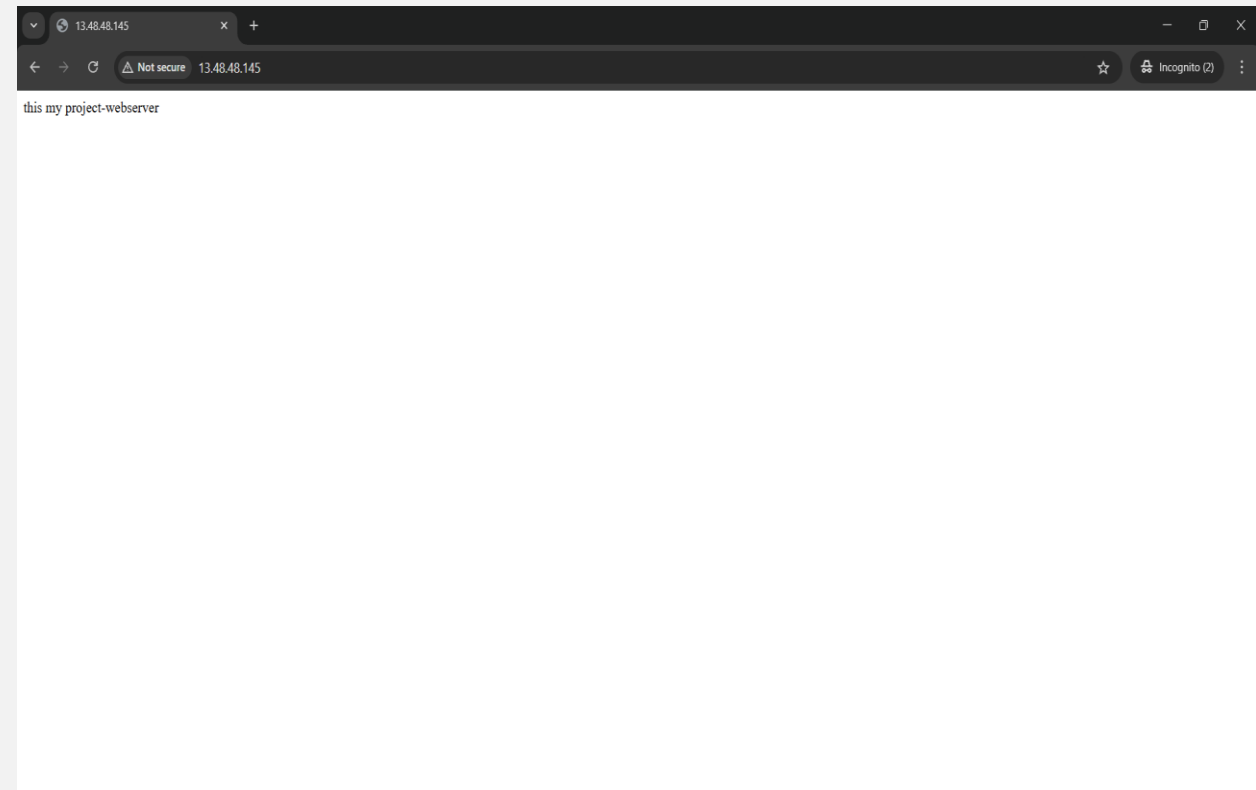
➤ We connect webserver 2 and start httpd and use public ip to display output



```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Feb 17 13:42:45 2026 from 13.48.4.203
[root@ip-172-31-30-97 ~]# systemctl start httpd
[root@ip-172-31-30-97 ~]#
```

i-02afc0b0ebaab7528 (webserver2)
PublicIPs: 13.48.48.145 PrivateIPs: 172.31.30.97



13.48.48.145

Not secure 13.48.48.145

this my project-webserver